MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(HONOURS) PHILOSOPHY EXAMINATION

III SEMESTER

OCTOBER 2025

Duration: 2 hrs Max.marks:70

MG3 DSC PHL 202 :INTRODUCTION TO SYMBOLIC LOGIC

Remember(k),understand(u),Apply(A),Analyse(An),Evaluated(E) create(C),skills (s)

Student should attempt at least one question fromeach course outcome to enhance their overall outcome attainability

PART A

ANSWER ANY 8 QUESTIONS. EACH QUESTION CARRIES 2 MARKS

1.state inference?[K,co1]

2. Distinguish between Truth and Validity.[u,co1]

3. What is meant by artifical language? [k,c01]

4. Write a note on disjunction [k,C01]

5.State the rule of Material Implication. If A, B and C are True Statements and X, Y and Z are False statements, state whether the following statement is true or false .[u,co2]

~ { ~ [~ (A ● ~ X) ● ~ A] ● ~ X }

6.Draw the Truth table of a Contradictory statement [k,Co2]

7. What is meant by Contingent Statement form? [k,c02]

8.What is meant by Logical Equivalence? [k,Co3]

10. Define Formal Proof of Validity. [k,co4]

11. State the Rule of Inference by which the Conclusion is deduced from the Premises [k,Co4]

A ● C .: (A ● C) v D

12. Write a short note on Disjunctive syllogism. [k,Co2]

[8x2=16]

PART B

ANSWER ANY SIX QUESTIONS. EACH QUESTION CARRIES 5 MARKS.

13. Give a short account on the relation between logic and language. [U,CO1]

14. Explain briefly three basic functions of language. [U,CO1]

15. Examine the difference between Truth Function and Truth Value. [An,CO2]

16. Write a short account on Material Equivalence [K,CO3]

17. Write short essay on Arguments . [K,CO1]

18. What is meant by Truth Table? How can we construct truth table for a 3 variable statement form. [K,CO2]

19. Use truth tables to determine the validity or invalidity of the following argument forms.

(p ⊃ q) ● (r ⊃ s) [An,CO3]

P v r

.: q v s

20. Elucidate structure of Formal Proof of Validity. [U,CO4]

21. Write 'Justification of each statement which is not a given on [U,CO4]

1. A ● B

2. (A v E) ⊃ F / .: A ● F

3. A

4. A v E

5. F

6. A ● F

22.Examine the shorter truth table method the validity/invalidity of the following argument[U,co4]

1. PƆ(q Ɔ r)

P Ɔ q/ .:pƆr

1. PƆ(q Ɔ r)
   rƆ ˜p/r Ɔ˜q [6x5=30]

Part C

Answer any two questions. Each question carries 12marks.

23. Explain the development of Symbolic logic from traditional logic. State the advantages of Symbolization.

24. Explain determination of Truth values in Truth functionally Compound Statements

25. Write an essay on different types of Statement forms

26. Prove the Invalidity of the following argument using Shorter Truth table Method

1. T ≡ U
   U ≡ (V ● W)
   V ≡ (T v X)
   T v X
   .: T ● X

2. (O v P) ⊃ Q

   Q ⊃ (P v R)

   (Q ⊃ O ~ S ⊃ P)

   (S ⊃ O) ⊃ ~R

   .: P ≡ Q

3. X ≡ (Y ⊃ Z)
   Y ≡ (~ X ● ~ Z)
   Z ≡ (X v ~ Y)
   Y
   .: XvZ

[2x12=24]